25901117 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBOT201, MG4DSCBBT200 - PLANT ANATOMY AND REPRODUCTIVE BOTANY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **10** questions

Each question carries **1** marks

1. Name the commonly used stain for microscopy. [K] / [CO1]
2. What is the use of iris diaphragm in a microscope? [K] / [CO1]
3. What is a microfibril in the ultrastructure of plant cell wall [U] / [CO2]
4. Name the substances that form the cementing material between adjacent plant cells. [K] / [CO2]
5. What is meant by bidirectional activity of cambium? [K] / [CO3]
6. Give one example of anomalous secondary thickening. [K] / [CO3]
7. What is the function of lenticels? [K] / [CO3]
8. Define the grain pattern in wood. [K] / [CO5]
9. What is male gametophyte? [K] / [CO6]

10. What is raphe? [K] / [CO6]

11. What is polyembryony? [K] / [CO6]

12. What is thalamus? [K] / [CO6]

[1x10 = 10]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

13. Describe the role of plasmodesmata in plant development ? [U] / [CO2]

14. How can cellulose be converted in to biofuels? [U] / [CO2]

15. List out the functions of external secretory tissues [K] / [CO1]

16. Briefly explain the characteristics of compression wood. [K] / [CO4]

17. With suitable examples, explain how anatomical characters used for identification of adulterance on crude drugs. [U] / [CO5]

18. How wood is modified for industrial use through physical methods? What are its advantage? [U] / [CO5]

19. Differentiate between albuminous seeds and exalbuminous seeds? [U] / [CO6]

20. Explain the structure of a typical female gametophyte. [U] / [CO6]

[5x6 = 30]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

21. Explain seasonal activity of cambium and formation of annual rings. Add a note on dendrochronology. [U] / [CO3]

22. Discuss the process and significance of syngamy and triple fusion [U] / [CO6]

[10x1 = 10]